Functions: Composite, Inverse & Logarithmic

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domain and range**

Find the domain of $f(x) = \frac{1}{x^2 - 9}$.

The denominator is zero when $x = \pm 3$. Domain: all real x except $x = 3$ and $x = -3$.

2 **[Foundational]** • **Target: Domain and range**

Find the range of $f(x) = -(x - 2)^2 + 5$.

Since $-(x - 2)^2 \leq 0$ for all x, $f(x) \leq 5$. Range: $f(x) \leq 5$.

3 **[Foundational]** • **Target: Composite functions**

Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.

$g(f(x)) = \left(\sqrt{x + 4}\right)^2 - 4 = (x + 4) - 4 = x$, valid for $x \geq -4$ (the domain of f).

4 **[Foundational]** • **Target: Verifying inverses**

Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x - 1}{5}$ are inverses by showing $f(g(x)) = x$.

$f(g(x)) = 5\left(\frac{x - 1}{5}\right) + 1 = (x - 1) + 1 = x$. Confirmed.

5 **[Foundational]** • **Target: Verifying inverses**

Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.

No. f is not one-to-one — it fails the horizontal line test, since for example $f(2) = f(-2) = 4$. An inverse function only exists if the domain is restricted (e.g. to $x \geq 0$).

6 **[Foundational]** • **Target: Logarithm laws**

Expand $\log_b\left(\frac{x^3 y}{z^2}\right)$ using the laws of logarithms.

$\log_b\left(\frac{x^3 y}{z^2}\right) = 3\log_b x + \log_b y - 2\log_b z$.

7 **[Foundational]** • **Target: Logarithm laws**

Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.

$2\log_5 x + 3\log_5 y - \log_5 z = \log_5\left(\frac{x^2 y^3}{z}\right)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Logarithm laws

Evaluate $\log_2 32 - \log_2 4$ without a calculator.

$\log_2\left(\frac{32}{4}\right) = \log_2 8 = 3$.

9 [Intermediate] • Target: Exponential equations and logs as inverses

The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

$81 = 3^4$, so $x = 4$. Since f and g are inverses, $g(f(16)) = 16$.

10 [Intermediate] • Target: Transformations of functions

The graph shows $f(x) = x^2$ (teal) and $g(x) = (x-2)^2 + 1$ (indigo). Describe the transformation that maps f onto g.

g is f shifted right 2 units and up 1 unit — a horizontal translation of $+2$ followed by a vertical translation of $+1$.

11 [Intermediate] • Target: Transformations of functions

If $h(x) = -f(x+3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h.

Shift f left 3 units (from $x+3$), then reflect the result over the x-axis (from the negative sign outside).

12 [Intermediate] • Target: A well-designed word problem: investment growth and doubling time

This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t, to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.

(a) $A(10) = 8000e^{0.45} \approx 8000(1.5683) \approx 12546$ dollars. (b) Doubling requires $e^{0.045t} = 2 \Rightarrow 0.045t = \ln 2 \Rightarrow t = \frac{\ln 2}{0.045} \approx 15.4$ years. (c) From $A = A_0 e^{rt}$: $\frac{A}{A_0} = e^{rt} \Rightarrow \ln\left(\frac{A}{A_0}\right) = rt \Rightarrow t(A) = \frac{1}{0.045}\ln\left(\frac{A}{A_0}\right)$.

13 [Intermediate] • Target: Verifying inverses

Given $f(x) = e^x$ and $g(x) = \ln(x+1)$, find the domain of $(g \circ f)(x)$ and evaluate $(g \circ f)(0)$.

Step 1: Formulate composite function: $(g \circ f)(x) = g(f(x)) = \ln(e^x + 1)$. Step 2: Analyze domain requirements: The argument of the natural log must be strictly positive: $e^x + 1 > 0$. Since $e^x > 0$ for all real numbers x, $e^x + 1 > 1 > 0$ holds universally. The domain is all real numbers: $\mathbf{(-\infty, \infty)}$. Step 3: Evaluate at $x = 0$: $(g \circ f)(0) = \ln(e^0 + 1) = \ln(1+1) = \ln\mathbf{(2)} \approx 0.693$.

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14 [Intermediate] • Target: Logarithm laws

The function $f(x) = \frac{4x-7}{2x+8}$ has horizontal asymptote $y = H$ and vertical asymptote $x = V$. Determine the value of $H + V$.

Step 1: Find horizontal asymptote $y = H$: Since degrees of numerator and denominator are both 1, the horizontal asymptote is the ratio of leading coefficients: $H = \frac{4}{2} = 2$. Step 2: Find vertical asymptote $x = V$: Set denominator to zero: $2x + 8 = 0 \Rightarrow 2x = -8 \Rightarrow V = -4$. Step 3: Compute sum: $H + V = 2 + (-4) = \mathbf{-2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

Write $y = \frac{3x-1}{2x+5}$. The domain of f is $x \neq -2.5$. Its horizontal asymptote is $y = 3/2$, so the range of f is $y \neq 1.5$. Swap x and y and solve for y: $x = \frac{3y-1}{2y+5} \Rightarrow x(2y+5) = 3y - 1$. $2xy + 5x = 3y - 1 \Rightarrow 2xy - 3y = -5x - 1$. $y(2x-3) = -(5x+1) \Rightarrow y = \frac{-5x-1}{2x-3} = \frac{5x+1}{3-2x}$. Inverse function: $f^{-1}(x) = \frac{5x+1}{3-2x}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (the range of f). Range of f^{-1}.

16 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 4.01 Lines, Triangles & Polygons

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 [Challenging] • Target: Transformations of functions • Cross-Lesson: 5.01 Trigonometry

An EST II Mathematics examination consists of 50 questions with an allocated testing time of 60 minutes. If a student completes the first 30 questions in 33 minutes, what is the maximum average time, in seconds, the student can spend on each of the remaining 20 questions?

Step 1: Compute total available time in seconds: $60 \text{ min} \times 60 \text{ s/min} = 3600 \text{ seconds}$. Step 2: Compute time spent on first 30 questions: $33 \text{ min} \times 60 \text{ s/min} = 1980 \text{ seconds}$. Step 3: Calculate remaining time: $3600 - 1980 = 1620 \text{ seconds}$. Step 4: Determine average time per remaining question: $\frac{1620 \text{ seconds}}{20 \text{ questions}} = 81 \text{ seconds per question}$.

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18 [Challenging] • Target: A well-designed word problem: investment growth and doubling time • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A trigonometric function has the formula $f(x) = -3\sin\left(2x - \frac{\pi}{2}\right) + 4$. State its: (a) amplitude, (b) period, (c) phase shift, and (d) range.

(a) Amplitude: $|A| = |-3| = \mathbf{3}$. (b) Period: Factor $B = 2$: $f(x) = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 4 \Rightarrow T = \frac{2\pi}{B} = \frac{2\pi}{2} = \boldsymbol{\pi}$. (c) Phase Shift: Shifted right by $\frac{\boldsymbol{\pi}}{\mathbf{4}}$. (d) Range: Minimum is $D - |A| = 4 - 3 = 1$; maximum is $D + |A| = 4 + 3 = 7$. Range is **[1, 7]**.

19 [Challenging] • Target: Domain and range • Cross-Lesson: 7.01 Solid Geometry & Similarity

Five test scores are 70, 75, 80, 85, 90. Find the mean and range.

Step 1: Compute arithmetic mean $\bar{x}$: $\bar{x} = \frac{70 + 75 + 80 + 85 + 90}{5} = \frac{400}{5} = \mathbf{80}$. Step 2: Compute range: Range = Maximum – Minimum = 90 – 70 = **20**.

20 [Challenging] • Target: Composite functions • Cross-Lesson: 8.01 Statistics & Probability

[Test format & strategy · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain and range

What value must be excluded from the domain of $f(x) = \frac{1}{3x + 3}$?

a. 3 **b.** **-1 ✓**

c. -4 **d.** 2

Denominator $= 0 \Rightarrow 3x + 3 = 0 \Rightarrow x = -1$, which must be excluded.

22 [Foundational] • Target: Composite functions

What is the smallest integer in the domain of $f(x) = \sqrt{2x + 2}$?

a. -8 **b.** -9

c. **-1 ✓** **d.** -10

Need $2x + 2 \geq 0 \Rightarrow x \geq -1$; smallest integer $= -1$.

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23 [Foundational] • Target: Inverse functions

Consider the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. If $f(g(x)) = \frac{1}{2x-1}$, find the value of k.

a. 3

b. 2

c. **1** ✓

d. -1

We are given the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. We are also given that $f(g(x)) = \frac{1}{2x-1}$. To find $f(g(x))$, we substitute $g(x)$ into $f(x)$: $f(g(x)) = f(3x + k) = \frac{1}{(3x+k)-2} = \frac{1}{3x+k-2}$. Now, we equate this to the given expression for $f(g(x))$: $\frac{1}{3x+k-2} = \frac{1}{2x-1}$. For these two fractions to be equal, their denominators must be equal: $3x + k - 2 = 2x - 1$. To solve for k, we need to rearrange the equation. It seems there might be a mistake in the.

24 [Foundational] • Target: Verifying inverses

Let $f(x) = \log_2(ax + b)$ and $g(x) = 3^x - 1$. Given that $f(g(x)) = x$ and the point $(2, 1)$ lies on the graph of $y = g(x)$, find the value of $a + b$.

a. **4** ✓

b. 3

c. 2

d. 1

We are given the function $f(x) = \log_2(ax + b)$. We are also given a function $g(x)$ and a point $(2, 1)$ that lies on its graph, and the condition $f(g(x)) = x$. The condition $f(g(x)) = x$ implies that f is the inverse function of g, i.e., $f(x) = g^{-1}(x)$. Let's first use the information about the point $(2, 1)$ and the structure of $g(x)$. If $g(x) = 2^x + k$ (assuming a general form for exponential functions with base 2 to match the logarithm base), and the point $(2, 1)$ lies on it, then: $g(2) = 1$.

25 [Foundational] • Target: Logarithm laws

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?

a. 1

b. **2** ✓

c. 3

d. 4

Let $y = f(x)$. Then $y = 2x + 3$. To find the inverse function $f^{-1}(y)$, we swap x and y and solve for y. So, $x = 2y + 3$. Subtract 3 from both sides: $x - 3 = 2y$. Divide by 2: $y = \frac{x-3}{2}$. Thus, $f^{-1}(x) = \frac{x-3}{2}$. We need to find $f^{-1}(7)$. Substitute $x = 7$ into the inverse function: $f^{-1}(7) = \frac{7-3}{2} = \frac{4}{2} = 2$.

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26 [Foundational] • Target: Solving logarithmic equations

Consider the functions $f(x) = 3x + 2$ and $g(x) = x - 5$. What is the value of $f(g(3))$?

a. **1** ✓

b. -4

c. 13

d. -10

To find $f(g(3))$, we first evaluate the inner function $g(3)$. $g(x) = x - 5$ $g(3) = 3 - 5 = -2$ Now we substitute the result of $g(3)$ into the outer function $f(x)$. $f(x) = 3x + 2$ $f(g(3)) = f(-2) = 3(-2) + 2$ $f(-2) = -6 + 2$ $f(-2) = -4$ Therefore, $f(g(3)) = -4$.

27 [Foundational] • Target: Exponential equations and logs as inverses

If $f(x) = -2x + 1$ and g is the inverse of f, find $g(f(-1))$.

a. 4

b. **-1** ✓

c. 2

d. -10

By definition of inverse functions, $g(f(x)) = x$ for all x, so $g(f(-1)) = -1$.

28 [Foundational] • Target: Transformations of functions

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8

b. 2

c. 4 ✓

d. 10

Applying Exponential & Logarithmic Functions principles, $2^{10/5} = 2^2 = 4$. Therefore, the verified correct option is 4.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: investment growth and doubling time

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. **0.0434 ✓**

We are given the exponential growth model $P(t) = P_0 e^{kt}$. We know that the initial population $P_0 = 500$ and that after 10 years, $P(10) = 1200$. We can substitute these values into the model: $1200 = 500e^{k \cdot 10}$ Divide both sides by 500: $\frac{1200}{500} = e^{10k}$ $2.4 = e^{10k}$ To solve for k, we take the natural logarithm of both sides: $\ln(2.4) = \ln(e^{10k})$ $\ln(2.4) = 10k$ Now, divide by 10: $k = \frac{\ln(2.4)}{10}$ Using a calculator, $\ln(2.4) \approx 0.8754687$. So.

30 [Intermediate] • Target: Domain and range

Consider the exponential growth model described by the function $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347 ✓**

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0 e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

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31 [Intermediate] • Target: Composite functions

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. **Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$** ✓

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (restricted principal output).

32 [Intermediate] • Target: Inverse functions

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. **Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$** ✓

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (restricted principal output).

33 [Intermediate] • Target: Verifying inverses

Which is the inverse of $f(x) = \log_2 x$?

a. $g(x) = \log_2 x$

b. $g(x) = 2x$

c. $g(x) = x^2$

d. $g(x) = 2^x$ ✓

Applying Exponential & Logarithmic Functions principles, Exponential base 2. Therefore, the verified correct option is $g(x) = 2^x$.

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34 [Intermediate] • Target: Logarithm laws

Use natural logs to solve $e^{x^2} = e^{3x+4}$.

a. $x = 4$ only

b. $x = -1$ only

c. $x = 3$

d. $x = 4$ **or** $x = -1$ ✓

$x^2 = 3x + 4 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0$.

35 [Intermediate] • Target: Solving logarithmic equations

Which function is the inverse of $f(x) = e^x$?

a. $g(x) = x^e$

b. $g(x) = \ln x$ ✓

c. $g(x) = 10^x$

d. $g(x) = \log x$

Natural log is inverse of natural exponential.

36 [Intermediate] • Target: Exponential equations and logs as inverses

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?

a. $-\frac{5}{2}$ ✓

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

To determine $f(-1)$, we first find the constants a, b, and c: 1. **Horizontal Asymptote:** For an exponential function of the form $f(x) = a \cdot b^x + c$, as the exponential term approaches zero, the function approaches its horizontal asymptote $y = c$. Given that the asymptote is $y = -4$, we have: $c = -4$ 2. **Use the Point $(0, -1)$:** Substitute $x = 0$ and $f(0) = -1$: $f(0) = a \cdot b^0 - 4 = -1$ $a(1) - 4 = -1 \Rightarrow a = 3$ 3. **Use the Point $(2, 8)$:** Substitute $x = 2$ and $f(2) = 8$.

37 [Intermediate] • Target: Transformations of functions

Find the range of the data set: $32, 35, 32, 11, 3, 3, 50$.

a. 21

b. 12

c. 18

d. **47** ✓

Range $= \max - \min = 50 - 3 = 47$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A well-designed word problem: investment growth and doubling time** • **Cross-Lesson: 6.01 Circles & Coordinate Geometry**

What is the sum of all real solutions to the equation $2^{\log_5(x^2)} - 6 \cdot x^{\log_5(2)} + 8 = 0$?

a. 30 ✓

b. 26

c. 6

d. 3

To solve the equation for real x, we first determine the domain and apply fundamental logarithmic identities. 1. **Domain:** The term $x^{\log_5(2)}$ requires $x > 0$ for the power to be well-defined over the real numbers. For $x > 0$, $\log_5(x^2)$ is also well-defined. 2. **Logarithmic and Exponential Identities:** Using the power rule for logarithms with $x > 0$: $\log_5(x^2) = 2\log_5(x)$ Thus, the first term can be rewritten as: $2^{\log_5(x^2)} = 2^{2\log_5(x)} = (2^{\log_5(x)})^2$ Next, recall t.

39 **[Challenging]** • **Target: Domain and range** • **Cross-Lesson: 7.01 Solid Geometry & Similarity**

In a right-angled triangle ABC, angle B is 90°. If $AB = 5$ and $BC = 12$, what is the length of the hypotenuse AC?

a. 10

b. 11

c. **13** ✓

d. 17

This problem involves finding the hypotenuse of a right-angled triangle given the lengths of the other two sides. We can use the Pythagorean theorem, which states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. Let a and b be the lengths of the two shorter sides, and c be the length of the hypotenuse. The Pythagorean theorem is given by: $a^2 + b^2 = c^2$. In triangle ABC, we have: $AB = a = 5$.

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40 [Challenging] • Target: Composite functions • Cross-Lesson: 8.01 Statistics & Probability

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. 20 ✓

b. 24

c. 32

d. 44

Step 1: Determine the vertical shift parameter c. For an exponential function in the form $f(x) = a \cdot b^x + c$, the term $a \cdot b^x$ approaches 0 as $x \to -\infty$ (for $b > 1$) or as $x \to \infty$ (for 00), we find $b = 2$. Step 4: Evaluate $f(3)$. The complete function is $f(x) = 3 \cdot 2^x - 4$. Evaluating at $x = 3$:
$f(3) = 3 \cdot 2^3 - 4 = 3(8) - 4 = 24 - 4 = 20$ Misconception Traps: - Choice B (24) results from computing $a \cdot b^3 = 3 \cdot 8 = 24$ but forgetting to subt.

41 [Challenging] • Target: Inverse functions • Cross-Lesson: 1.01 Advanced Algebra

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19 ✓

b. 31

c. 16

d. 11

For an exponential function of the form $f(x) = a \cdot 2^x + k$, as $x \to -\infty$, $2^x \to 0$, which gives the horizontal asymptote $y = k$. Since the horizontal asymptote is given as $y = 3$, we have: $k = 3$ Next, the graph passes through $(0, 7)$, so $f(0) = 7$: $a \cdot 2^0 + 3 = 7$ Since $2^0 = 1$, this simplifies to: $a + 3 = 7 \Rightarrow a = 4$ Thus, the function is $f(x) = 4 \cdot 2^x + 3$. Evaluating at $x = 2$: $f(2) = 4 \cdot 2^2 + 3 = 4 \cdot 4 + 3 = 16 + 3 = 19$ Common errors: - Se.

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42 [Challenging] • Target: Verifying inverses • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

In a right-angled triangle ABC, where angle B is 90 degrees, if AB = 3 and BC = 4, what is the length of the hypotenuse AC?

a. 7

b. 5 ✓

c. 12

d. 25

This question involves a right-angled triangle and requires the application of the Pythagorean theorem. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In triangle ABC, AC is the hypotenuse. According to the theorem: $AC^2 = AB^2 + BC^2$ Substituting the given values: $AC^2 = 3^2 + 4^2$ $AC^2 = 9 + 16$ $AC^2 = 25$ To find the length of AC, we take the square root of both sides.

43 [Challenging] • Target: Logarithm laws • Cross-Lesson: 4.01 Lines, Triangles & Polygons

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$ ✓

b. $y = 0$

c. $y = 3$

d. $x = -1$

An exponential function in the standard form $f(x) = a \cdot b^{x-h} + k$ (with base $b > 1$) has a horizontal asymptote determined by its vertical shift, $y = k$. As $x \to -\infty$, the exponential power $2^{x+1} \to 0$. Substituting this into the function yields: $\lim_{x \to -\infty} f(x) = 3(0) - 5 = -5$ Thus, the horizontal asymptote is the horizontal line $y = -5$. Common Misconceptions: - $y = 0$: Assumes the asymptote is always the x-axis, forgetting the vertical shift by -5..

44 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 5.01 Trigonometry

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. **a = 4, b = 3** ✓

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(3, 108)$. Substituting these points into the function equation, we get two equations: 1) $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2) $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this expression for a into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12 \cdot b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b must be positive for an exponen.

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45 [Challenging] • Target: Exponential equations and logs as inverses • Cross-Lesson: 6.01 Circles & Coordinate Geometry

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. **4** ✓

d. 6

Since the graph of $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$, we can set up a system of two equations: 1. $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2. $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b is the base of an exponential function, $b > 0$. Thus, $b = 3$. Now, substitute the value of b back into equ.

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